

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Cost = 108×16 or 108×0.16 (1) substitution Cost = £17.28 (1) answer Accept £17 or £17.00	1	1		2	2	
		(iii)		Running cost of TV 2 for 10 years = $\text{£}17.28$ (ecf) $\times 10 = \text{£}172.80$ (1) Accept £170 or £172 or £173 Running cost of TV 4 for 10 years = $172 \times 10 \times 0.16 = \text{£}275.20$ (1) Accept £275 TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) Alternative: Annual savings from using TV 2 = $(172 - 108) \times 0.16 = \text{£}10.24$ (1) Running cost = $\text{£}10.24 \times 10 = \text{£}102.40$ (1) OR Difference in units over 10 years $(172 - 108) \times 10 = 640$ (1) Difference in running cost = $640 \times 0.16 = \text{£}102.40$ (1) 3rd mark - TV 4 costs £102.40 more to run but it is £200 cheaper to buy so Sarah is right (1) Alternative: Total cost of TV 2 = $\text{£}172.80$ ecf (1) + $\text{£}1\,000 = \text{£}1\,172.80$ (1) Total cost of TV 4 = $\text{£}1\,075.20$ so cheaper so Sarah is right (1) OR Total cost of TV 4 = $\text{£}275.20$ (1) + $\text{£}800 = \text{£}1\,075.20$ (1) Total cost of TV 2 = $\text{£}1\,172.80$ so more expensive so Sarah is right (1) Alternative: Annual savings from using TV 2 = $(172 - 108) \times 0.16 = \text{£}10.24$ (1) Payback time = $\frac{200}{10.24}$ (1) = 19.5 years which is longer than 10 years so Sarah is right (1)			3	3	2	